TEJAS 200 – PREDICTED PAPER 01

JEE Main 2027 pattern — Mathematics, Physics, Chemistry

Time: 3 hours **Maximum marks:** 300 **Questions:** 75 (25 per subject)

Instructions

- Each subject has Section A (20 multiple-choice questions, exactly one correct option) and Section B (5 numerical-value questions; the answer is an integer). All 75 questions are compulsory.
- Marking: +4 for a correct answer, −1 for a wrong answer, 0 if not attempted (Section A and Section B).
- Sit it like the real exam: one sitting, 180 minutes, no breaks. Note the time you start each subject.
- After the paper, for every question note: concept recognised? (Y/N), correct? (Y/N), time taken.

MATHEMATICS – SECTION A (Q1–Q20) Multiple choice, one correct option

Q1. Let the foci of a hyperbola H be $S(-1,3)$ and $S'(-1,-1)$, and let H pass through the point $P(11,8)$. If l is the length of the latus rectum of H and d is the distance of P from the directrix of H corresponding to the focus S, then $l+2d$ is equal to :

(1) 16 **(2)** 19 **(3)** 21 **(4)** 23

Q2. The sum of the series $\frac{1^3}{2}+\frac{1^3+2^3}{2+4}+\frac{1^3+2^3+3^3}{2+4+6}+\cdots+\frac{1^3+2^3+\cdots+15^3}{2+4+\cdots+30}$ is equal to :

(1) 280 **(2)** 408 **(3)** 680 **(4)** 340

Q3. Let $\vec{a}=4\hat{i}+\hat{k}$ and $\vec{b}=\hat{j}+\hat{k}$. Let $\vec{c}=c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$, with $c_1c_2c_3\neq 0$, be a vector such that $\vec{a}\times\vec{c}=2(\vec{c}\times\vec{b})$. If the circle $x^2+y^2+c_1x+c_2y+c_3=0$ touches the y-axis and meets the x-axis at the points A and B, then the length AB is equal to :

(1) $6\sqrt{3}$ **(2)** $2\sqrt{3}$ **(3)** $6\sqrt{15}$ **(4)** $8\sqrt{15}$

Q4. Let $\vec{d}$ be a vector such that $\vec{d}\times(\hat{i}+\hat{j}-\hat{k})=-\hat{i}+\hat{j}$ and $\vec{d}\cdot\hat{k}=-1$. A line L, parallel to $\vec{d}$, meets the lines $L_1:\frac{x}{2}=\frac{y}{2}=\frac{z}{1}$ and $L_2:\frac{x}{-2}=\frac{y-3}{1}=\frac{z+3}{2}$ at the points P and Q respectively. Then the length PQ is equal to :

(1) 4 **(2)** 6 **(3)** 9 **(4)** 3

Q5. Let $L_1:3x-4y+5=0$ and $L_2:5x+12y-29=0$ be two lines. If the bisector of the obtuse angle between L_1 and L_2 meets the coordinate axes at the points P and Q, then the area (in sq. units) of the triangle OPQ, O being the origin, is :

(1) $\frac{225}{16}$ **(2)** $\frac{25}{4}$ **(3)** $\frac{289}{16}$ **(4)** $\frac{9}{2}$

Q6. A bag contains 2 white and 4 black balls. A ball is drawn at random from the bag 5 times with replacement. The probability that a white ball is drawn at least 3 times is :

(1) $\frac{51}{243}$ **(2)** $\frac{40}{243}$ **(3)** $\frac{232}{243}$ **(4)** $\frac{50}{243}$

Q7. Let $\omega\neq 1$ be a cube root of unity. Then $\sum_{k=1}^{30} k\left(\omega^k+\omega^{2k}\right)^3$ is equal to :

(1) 750 **(2)** 1620 **(3)** 1020 **(4)** 1320

Q8. For a natural number n, let $D(n) = \begin{vmatrix} {}^{n}C_0 & {}^{n}C_2 & {}^{n}C_3 \\ {}^{n+1}C_0 & {}^{n+1}C_2 & {}^{n+1}C_3 \\ {}^{n+2}C_0 & {}^{n+2}C_2 & {}^{n+2}C_3 \end{vmatrix}$, where ${}^{m}C_r = 0$ for $r > m$. If $\sum_{n=1}^{k} D(n)$ is equal to the coefficient of x in the expansion of $\left(x + \frac{1}{x^2}\right)^{10}$, then k is equal to :

(1) 9 **(2)** 10 **(3)** 7 **(4)** 8

Q9. If all the words (with or without meaning) formed by using all the letters of the word MADRAS are arranged as in a dictionary, then the position of the word MADRAS in this list is :

(1) 189 **(2)** 249 **(3)** 129 **(4)** 188

Q10. Let k arithmetic means $A_1, A_2, \ldots, A_k$ be inserted between 2 and 38 such that $A_2 : A_{k-1} = 1 : 4$. Then the sum $A_1 + A_2 + \cdots + A_k$ is equal to :

(1) 240 **(2)** 220 **(3)** 260 **(4)** 200

Q11. Let $L : \frac{x-3}{2} = \frac{y-1}{1} = \frac{z}{-2}$ be a line and let $\vec{u}$ be a vector such that $\vec{u} \times (\hat{i} + \hat{j}) = -\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{u} \cdot \hat{j} = -1$. If the line through the point $P(\lambda, 5, -5)$, parallel to $\vec{u}$, meets L at the point Q, then PQ^2 is equal to :

(1) 26 **(2)** 43 **(3)** 11 **(4)** 27

Q12. The sum of the infinite series $\frac{1}{6} + \frac{5}{36} + \frac{19}{216} + \frac{65}{1296} + \ldots$ is equal to :

(1) $\frac{3}{4}$ **(2)** $\frac{1}{3}$ **(3)** $\frac{1}{2}$ **(4)** 1

Q13. Let $I_n = \int_0^{\pi/4} \tan^n x \, dx$, $n \in \mathbb{N}$. Then $I_3 - I_5$ is equal to :

(1) $\frac{3}{4} - \log_e 2$ **(2)** $\log_e 2 - \frac{1}{2}$ **(3)** $\frac{1}{2}\log_e 2 - \frac{1}{4}$ **(4)** $1 - \log_e 2$

Q14. Let $f(x) = \sin^{-1}\left(\frac{2x+[x]}{5}\right)$, where $[x]$ denotes the greatest integer less than or equal to x. If the domain of f is $[\alpha, \beta)$, then $\alpha^2 + \beta^2$ is equal to :

(1) 5 **(2)** $\frac{25}{4}$ **(3)** $\frac{13}{4}$ **(4)** $\frac{50}{9}$

Q15. Let E be the ellipse whose foci are $(\pm 3\sqrt{3}, 0)$ and whose minor axis is of length 6. If l is the length of the chord of E whose midpoint is the point $(4, 2)$, then l^2 is equal to :

(1) 8 **(2)** 40 **(3)** 10 **(4)** 90

Q16. Let $f(x)$ be a polynomial of degree 3 having local extrema at $x = 2$ and $x = 4$, and let $\lim_{x\to 0} \frac{f(x)-3}{x} = 24$. Then the local minimum value of f is equal to :

(1) 23 **(2)** 21 **(3)** 3 **(4)** 19

Q17. Let F be the focus of the parabola $x^2 - 4x + 8y - 4 = 0$. Then the minimum distance between F and a point on the circle $x^2 + y^2 - 28x + 12y + 207 = 0$ is :

(1) 10 **(2)** 8 **(3)** 18 **(4)** 13

Q18. Let $y = y(x)$ be the solution of the differential equation $x\frac{dy}{dx} + \sin 2y = 4x^2 \cos^2 y$, $x > 0$, with $y(1) = \frac{3\pi}{4}$. Then $\sin\left(y(\sqrt{2})\right) + \cos\left(y(\sqrt{2})\right)$ is equal to :

(1) $-\sqrt{2}$ **(2)** $\sqrt{2}$ **(3)** 0 **(4)** $-\frac{3}{\sqrt{5}}$

Q19. The number of integral values of k for which the equation $8\sin^2 x + 4\cos x = k$ has at least one real solution is :

(1) 9 **(2)** 12 **(3)** 13 **(4)** 17

Q20. Let T_1 be the equilateral triangle with one vertex at $A(1,1)$ and the opposite side lying on the line $3x+4y+8=0$. A triangle T_2 is formed by joining the midpoints of the sides of T_1, a triangle T_3 by joining the midpoints of the sides of T_2, and so on. Then the sum of the areas (in sq. units) of all the triangles $T_1, T_2, T_3, \ldots$ is :

(1) $3\sqrt{3}$ **(2)** $6\sqrt{3}$ **(3)** $\sqrt{3}$ **(4)** $4\sqrt{3}$

MATHEMATICS – SECTION B (Q21–Q25) Numerical value (integer answer)

Q21. Let $A=\{1,2,3\}$ and $B=\{x\in\mathbb{N} : \log_2 x \text{ is a positive integer and } x<20\}$. Then the number of onto functions $f: B\to A$ such that $f(2)\neq f(4)$ is _____.

Q22. If the sum of the binomial coefficients in the expansion of $\left(x^3-\frac{2}{x^2}\right)^n$ is 1024, then the term independent of x in this expansion is _____.

Q23. The value of $\pi\sum_{r=1}^{30}\int_0^r (x+1)\,|\cos\pi x|\,dx$ is equal to _____.

Q24. Let $f(x)=\begin{cases} ax^2+bx, & x\le 1 \\ a^2x-b, & x>1 \end{cases}$, where $a>0$, be differentiable at $x=1$. If A is the area of the region bounded by the curve $y=f(x)$, the x-axis and the lines $x=-1$ and $x=2$, then $2A$ is equal to _____.

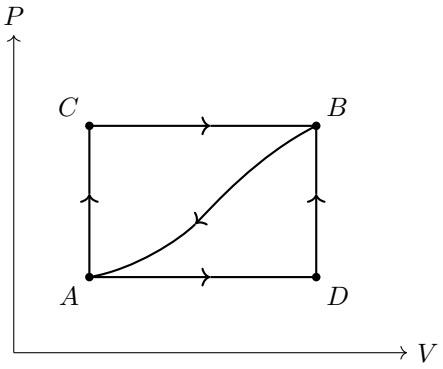
Q25. If the sum of all the solutions of the equation $\sin 3x+\sin x=\sin 2x$ in the interval $[0,2\pi]$ is $k\pi$, then k is equal to _____.

PHYSICS – SECTION A (Q26–Q45) Multiple choice, one correct option

Q26. A projectile of mass 500 g is projected from the ground with a speed of 20 m s^{-1} at an angle of 60° with the horizontal. The work done by the gravitational force on the projectile from the instant of projection up to the instant when its velocity makes an angle of 45° with the horizontal is (take $g = 10$ m s^{-2}):

(1) 50 J **(2)** -75 J **(3)** -50 J **(4)** -25 J

Q27. An ideal gas is taken from state A to state B along the path ACB shown in the P-V diagram. It absorbs 96 J of heat and does 36 J of work. Along the path ADB, the work done by the gas is 16 J. When the gas is brought back from B to A along the curved path, 24 J of work is done on the gas. The heat absorbed by the gas along ADB and the heat released by the gas along the curved path are, respectively:



(1) 76 J and 84 J **(2)** 76 J and 36 J **(3)** 44 J and 84 J **(4)** 44 J and 36 J

Q28. A block of mass 8 kg rests on a smooth horizontal surface and is attached to a horizontal spring of spring constant 1000 N m^{-1}. A second block of mass 2 kg is placed on top of the first block. The coefficient of friction between the two blocks is 0.5. The maximum amplitude of simple harmonic motion of the system for which the upper block does not slip on the lower block is (take $g = 10$ m s^{-2}):

(1) 4 cm **(2)** 5 cm **(3)** 1 cm **(4)** 10 cm

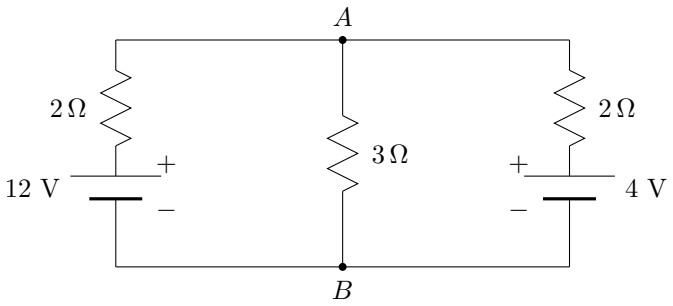
Q29. The energy E of a particle at position x moving with speed v is given by $E = Ax^2 + \dfrac{B}{x} + Cv$, where A, B and C are constants. The dimensional formula of $\dfrac{BC}{A}$ is:

(1) $[\mathrm{M\,L^4\,T^{-3}}]$ **(2)** $[\mathrm{M^{-1}\,L^{-4}\,T}]$ **(3)** $[\mathrm{M^2\,L^4\,T^{-3}}]$ **(4)** $[\mathrm{M\,L^4\,T^{-1}}]$

Q30. Two planets A and B are spheres of uniform density. The radius of B is three times the radius of A, and the mean density of B is four times that of A. If a satellite revolving close to the surface of A has a period T, then the period of a satellite revolving close to the surface of B is:

(1) $T/2$ **(2)** $2T$ **(3)** T **(4)** $\dfrac{\sqrt{3}}{2}T$

Q31. In the circuit shown in the figure, the cells have negligible internal resistance. The heat produced in the $3\,\Omega$ resistor in 10 s is:



(1) 270 J **(2)** 480 J **(3)** 120 J **(4)** 60 J

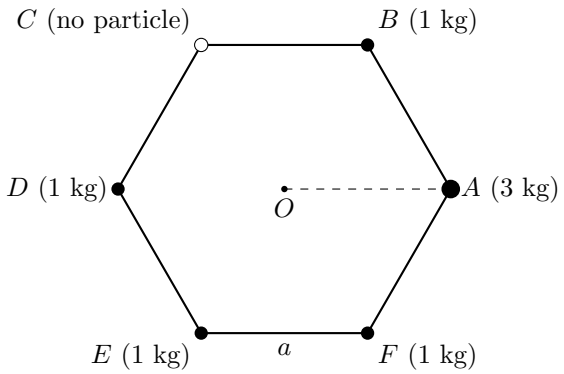
Q32. Two large parallel plates P and Q, placed 1 cm apart, are kept at potentials $+150$ V and -50 V respectively. A uniform magnetic field of 0.05 T exists in the region between the plates, parallel to the plates. A particle of mass 1.6×10^{-27} kg and charge $+1.6 \times 10^{-19}$ C is released from rest close to plate P. Neglecting gravity, the speed with which the particle strikes plate Q is:

(1) $\sqrt{2} \times 10^5$ m s^{-1} **(2)** 2×10^5 m s^{-1} **(3)** $\sqrt{3} \times 10^5$ m s^{-1} **(4)** 4×10^5 m s^{-1}

Q33. A cylindrical vessel standing on a block of height 80 cm contains water filled up to a height of 125 cm. A small hole is made in the side wall of the vessel at a height of 45 cm above the base of the vessel. The horizontal distance from the hole at which the jet of water coming out of the hole strikes the ground is equal to : (take $g = 10$ m s^{-2})

(1) 1.2 m **(2)** 2.5 m **(3)** 1.6 m **(4)** 2 m

Q34. Particles of masses 3 kg, 1 kg, 1 kg, 1 kg and 1 kg are placed at the vertices A, B, D, E and F respectively of a regular hexagon $ABCDEF$ of side a, and there is no particle at the vertex C, as shown in the figure. The distance of the centre of mass of the system from the centre O of the hexagon is:



(1) $\dfrac{\sqrt{7}\,a}{7}$ **(2)** $\dfrac{\sqrt{3}\,a}{7}$ **(3)** $\dfrac{3a}{7}$ **(4)** $\dfrac{\sqrt{7}\,a}{6}$

Q35. Two blocks of masses 3 kg and 1 kg are connected by a light inextensible string passing over a pulley which is a uniform disc of mass 2 kg mounted on a frictionless horizontal axle. The string does not slip on the pulley. At $t = 0$, the 3 kg block is moving vertically upwards with a speed of 10 m s^{-1}. The distance travelled by the 3 kg block in the first 4 s is (take $g = 10$ m s^{-2}):

(1) 8 m **(2)** 17 m **(3)** 20 m **(4)** 12.5 m

Q36. A composite rod is made by joining end to end a brass rod of length 60 cm and a steel rod of length 40 cm at 20 °C. The coefficients of linear expansion of brass and steel are 2.0×10^{-5} °C^{-1} and 1.2×10^{-5} °C^{-1} respectively. The length of the composite rod at 120 °C is:

(1) 100.152 cm **(2)** 100.200 cm **(3)** 100.168 cm **(4)** 101.680 cm

Q37. If force F, velocity v and time T are taken as the fundamental quantities, then match List-I with List-II.

List-I: (A) Mass; (B) Energy; (C) Density; (D) Linear momentum.

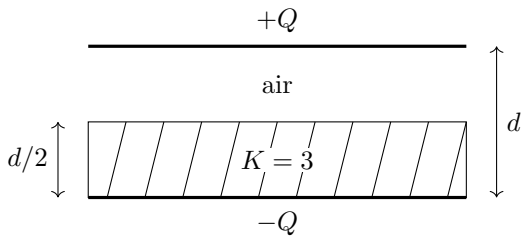
List-II: (I) $[F\,v\,T]$; (II) $[F\,T]$; (III) $[F\,v^{-1}\,T]$; (IV) $[F\,v^{-4}\,T^{-2}]$.

Choose the correct answer from the options given below:

(1) (A)-(I), (B)-(III), (C)-(IV), (D)-(II) **(2)** (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(3) (A)-(III), (B)-(I), (C)-(II), (D)-(IV) **(4)** (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

Q38. A parallel plate capacitor with air between its plates is charged by a battery and then disconnected from it. A dielectric slab of dielectric constant $K = 3$ and thickness equal to half the plate separation is now inserted so that it covers the entire plate area, as shown in the figure. The ratio of the energy stored in the capacitor after the insertion of the slab to that before is:



(1) $\frac{3}{2}$ **(2)** $\frac{2}{3}$ **(3)** $\frac{1}{2}$ **(4)** $\frac{1}{3}$

Q39. A large piece of ice at 0 °C is put into 300 g of water at 40 °C contained in a vessel of negligible heat capacity. When thermal equilibrium is reached, some ice is still left unmelted. The mass of ice that has melted is (latent heat of fusion of ice = 80 cal g^{-1}, specific heat of water = 1 cal g^{-1} °C^{-1}):

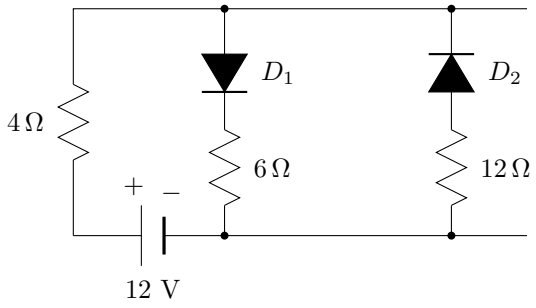
(1) 150 g **(2)** 600 g **(3)** 100 g **(4)** 450 g

Q40. A nucleus of mass number 224, initially at rest, decays by emitting an α-particle. The Q-value of the decay is 6.16 MeV. Treating the nuclear masses as proportional to the mass numbers, the kinetic energies of the α-particle and of the daughter nucleus are, respectively:

(1) 0.110 MeV and 6.050 MeV **(2)** 3.080 MeV and 3.080 MeV

(3) 6.050 MeV and 0.110 MeV **(4)** 6.048 MeV and 0.112 MeV

Q41. In the circuit shown in the figure, D_1 and D_2 are ideal diodes and the battery has negligible internal resistance. The current drawn from the battery is:



(1) 1.5 A **(2)** 0.75 A **(3)** 2 A **(4)** 1.2 A

Q42. Two soap bubbles of radii 4 cm and 6 cm, blown from the same soap solution, come in contact and share a common surface. The radius of curvature of the common surface is:

(1) 12 cm **(2)** 2.4 cm **(3)** 2 cm **(4)** 24 cm

Q43. A conducting rod of length 0.5 m and resistance $1\,\Omega$ slides with a constant velocity of 12 m s^{-1} on two long, parallel, smooth conducting rails of negligible resistance, in a uniform magnetic field of 0.5 T perpendicular to the plane of the rails. The rails are connected at one end through a $3\,\Omega$ resistor and at the other end through a $6\,\Omega$ resistor. The potential difference between the ends of the rod is:

(1) 3 V **(2)** 2 V **(3)** 2.7 V **(4)** 1 V

Q44. A uniform wooden cylinder of length 2.5 m and density 640 kg m^{-3} floats vertically in water of density 1000 kg m^{-3}. It is pushed down slightly and released. The time period of the resulting vertical oscillations is (take $g = 10$ m s^{-2}):

(1) π s **(2)** 0.4π s **(3)** 0.6π s **(4)** 0.8π s

Q45. The electric field of a plane electromagnetic wave travelling along the positive x-direction is given by $\vec{E} = 90\sin(kx - \omega t)\,(-\hat{j})$ V m^{-1}. Taking $c = 3 \times 10^8$ m s^{-1}, the magnetic field of the wave is:

(1) $\vec{B} = 3 \times 10^{-7}\sin(kx - \omega t)\,\hat{k}$ T **(2)** $\vec{B} = 2.7 \times 10^{10}\sin(kx - \omega t)\,(-\hat{k})$ T

(3) $\vec{B} = 3 \times 10^{-7}\sin(kx - \omega t)\,(-\hat{k})$ T **(4)** $\vec{B} = 3 \times 10^{-7}\sin(kx - \omega t)\,\hat{i}$ T

PHYSICS – SECTION B (Q46–Q50) Numerical value (integer answer)

Q46. A circular coil of 100 turns and area 25 cm^2 has a resistance of $10\,\Omega$. It is placed with its plane perpendicular to a magnetic field which varies with time as $B = 0.2\sin(200t)$ T, where t is in seconds. The average power dissipated in the coil is _____ W.

Q47. A series LCR circuit with $R = 40\,\Omega$ and $L = \dfrac{0.8}{\pi}$ H is connected to a 220 V, 50 Hz AC supply. The current in the circuit lags behind the applied voltage and the power factor of the circuit is 0.8. The capacitive reactance of the circuit is _____ Ω.

Q48. A thin convex lens of focal length 12 cm forms a virtual image three times the size of the object. The distance between the object and the image is _____ cm.

Q49. A compound microscope has an objective of focal length 0.5 cm and an eyepiece of focal length 2.5 cm, with a tube length of 15 cm. In normal adjustment, the magnifying power of the microscope (take $D = 25$ cm) is _____.

Q50. The objective of an astronomical telescope has a diameter of 11 cm. For light of wavelength 550 nm, the minimum angular separation between two distant point objects that the telescope can just resolve is _____ $\times 10^{-7}$ rad.

CHEMISTRY – SECTION A (Q51–Q70) Multiple choice, one correct option

Q51. Which of the following statements are correct?

(A) Chlorobenzene is much less reactive than chloroethane towards nucleophilic substitution by aqueous KOH.

(B) Among the isomeric alkyl bromides C_4H_9Br, 2-bromo-2-methylpropane has the highest boiling point.

(C) 2-Bromobutane, on heating with alcoholic KOH, gives but-2-ene as the major product.

(D) Ethyl fluoride can be prepared by heating ethyl chloride with NaF in dry acetone.

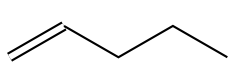
(E) For a given alkyl group, the boiling points of the halides follow the order R−I > R−Br > R−Cl > R−F.

Choose the correct answer from the options given below:

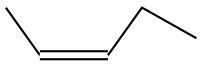
(1) (A), (C), (D) and (E) only **(2)** (A), (C) and (E) only

(3) (A), (B), (C) and (E) only **(4)** (C) and (E) only

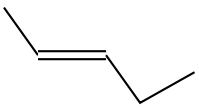
Q52. The correct order of stability of the alkenes (P), (Q), (R) and (S) shown in the figure is



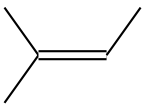
(P)



(Q)



(R)



(S)

(1) S > R > Q > P **(2)** S > Q > R > P **(3)** P > Q > R > S **(4)** S > R = Q > P

Q53. Given below are two statements:

Statement I: 3-Methylbut-1-ene reacts with HBr in the absence of peroxide to give 2-bromo-3-methylbutane as the major product.

Statement II: 3-Methylbut-1-ene, on treatment with Br_2 in CCl_4, then with excess $NaNH_2$ followed by water, and finally with $HgSO_4$ in dilute H_2SO_4, gives 3-methylbutan-2-one.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true

Q54. The crystal field stabilisation energy (CFSE) of the tetrahedral complex $[CoCl_4]^{2-}$, expressed in terms of Δ_o, is

(Given: $\Delta_t = \frac{4}{9}\Delta_o$)

(1) $-\frac{6}{5}\Delta_o$ **(2)** $-\frac{16}{45}\Delta_o$ **(3)** $-\frac{8}{15}\Delta_o$ **(4)** $-\frac{4}{5}\Delta_o$

Q55. Among N_2, O_2, F_2 and N_2^-, the species for which the removal of one electron decreases the bond order but increases the number of unpaired electrons is

(1) O_2 **(2)** N_2 **(3)** N_2^- **(4)** F_2

Q56. The structures (A), (B), (C) and (D) shown in the figure are resonating structures of prop-2-enal (CH_2=CH−CHO). The least stable resonating structure is

(A) $H_2C=CH-CH=O$

(B) $H_2C=CH-CH^+-O^-$

(C) $H_2C^+-CH=CH-O^-$

(D) $H_2C^--CH=CH-O^+$

(1) (A) **(2)** (B) **(3)** (C) **(4)** (D)

Q57. The gas-phase decomposition $A(g) \rightarrow 2B(g) + C(g)$ is a first-order reaction. Starting with pure A at an initial pressure of 400 mm Hg in a closed vessel at constant temperature, the total pressure becomes 600 mm Hg after 10 minutes. The rate constant of the reaction is

(Given: $\log 2 = 0.30$, $\log 3 = 0.48$)

(1) 2.76×10^{-2} min^{-1} **(2)** 1.84×10^{-2} min^{-1} **(3)** 4.15×10^{-2} min^{-1} **(4)** 6.91×10^{-2} min^{-1}

Q58. In the Liebig combustion analysis of 0.25 g of an organic compound, the increase in the mass of the U-tube containing anhydrous $CaCl_2$ was 225 mg and that of the bulb containing KOH solution was 660 mg. The mass of oxygen present in the sample is

(1) 70 mg **(2)** 57.5 mg **(3)** 45 mg **(4)** 4.5 mg

Q59. Consider the following amino acids:

Valine, Glycine, Lysine, Methionine, Proline

The number of essential amino acids among the above is :

(1) 2 **(2)** 3 **(3)** 4 **(4)** 5

Q60. Which of the following statements about lanthanoids are correct?

(A) Ce^{4+} is a strong oxidising agent in aqueous solution and gets reduced to Ce^{3+}.

(B) Eu^{2+} is a strong reducing agent and changes to the +3 state.

(C) Tb^{4+} and Gd^{3+} have the same number of 4f electrons.

(D) Tb^{3+} has a half-filled 4f subshell.

(E) As a consequence of lanthanoid contraction, the atomic radii of Zr and Hf are almost identical.

Choose the correct answer from the options given below:

(1) (A), (B), (C) and (E) only **(2)** (A), (B), (D) and (E) only

(3) (B), (C) and (E) only **(4)** (A), (C), (D) and (E) only

Q61. Which of the following statements is correct?

(1) The bond dissociation enthalpy of F_2 is higher than that of Cl_2.

(2) Fluorine stabilises the highest oxidation state of manganese more effectively than oxygen does.

(3) The oxidation state of oxygen in OF_2 is −2.

(4) The highest fluoride of manganese is MnF_4, whereas its highest oxide is Mn_2O_7.

Q62. At 298 K, the correct order of molar heat capacities at constant pressure ($C_{p,m}$) of $H_2O(l)$, $N_2(g)$, Al(s) and Ar(g) is

(1) $H_2O(l) > Al(s) > N_2(g) > Ar(g)$ **(2)** $N_2(g) > Ar(g) > H_2O(l) > Al(s)$

(3) $H_2O(l) > N_2(g) > Al(s) > Ar(g)$ **(4)** $Al(s) > H_2O(l) > N_2(g) > Ar(g)$

Q63. The pH of a 0.1 M aqueous solution of sodium cyanide at 298 K is

(Given: K_a of HCN $= 1.0 \times 10^{-10}$)

(1) 11.5 **(2)** 2.5 **(3)** 12.0 **(4)** 11.0

Q64. The IUPAC name of the complex $K[Cr(en)(C_2O_4)_2]$ (en = ethane-1,2-diamine) is

(1) Potassium dioxalato(ethane-1,2-diamine)chromate(III)

(2) Potassium (ethane-1,2-diamine)dioxalatochromate(III)

(3) Potassium (ethane-1,2-diamine)dioxalatochromate(IV)

(4) Potassium (ethane-1,2-diamine)dioxalatochromium(III)

Q65. Given below are two statements:

Statement I: Steam distillation is used to separate substances which are steam volatile and miscible with water.

Statement II: Solutions showing a large positive deviation from Raoult's law form minimum boiling azeotropes.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

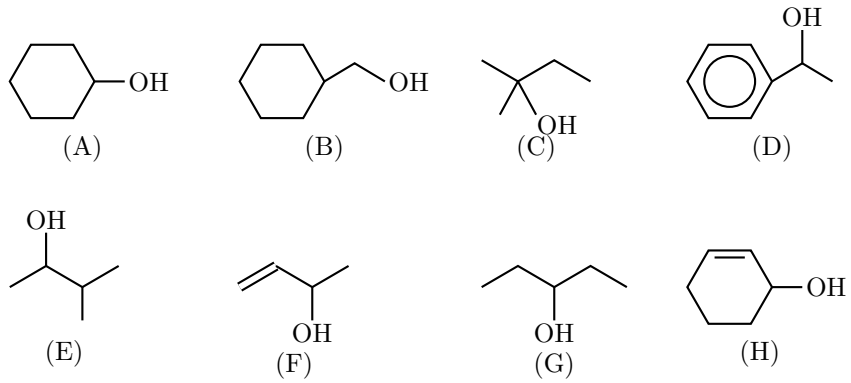
(4) Statement I is false but Statement II is true

Q66. The wavelength of the series limit of the Balmer series of He^+ is

(Given: $R_H = 1.1 \times 10^7$ m^{-1})

(1) 364 nm **(2)** 182 nm **(3)** 90.9 nm **(4)** 9.09 nm

Q67. The number of structures among (A)-(H) shown in the figure that are secondary alcohols and are also chiral is



(1) 4 **(2)** 6 **(3)** 3 **(4)** 5

Q68. According to the electrochemical theory of rusting of iron, the reaction occurring at the cathodic area on the surface of iron is

(1) $Fe \rightarrow Fe^{2+} + 2e^-$ **(2)** $2H^+ + 2e^- \rightarrow H_2$

(3) $Fe^{2+} \rightarrow Fe^{3+} + e^-$ **(4)** $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$

Q69. When 4.5 g of a non-volatile, non-electrolyte solute is dissolved in 147 g of acetic acid, the boiling point of the solvent is raised by 1.02 K. The relative lowering of vapour pressure of this solution is

(Given: K_b of acetic acid = 3.0 K kg mol^{-1}, molar mass of acetic acid = 60 g mol^{-1})

(1) 0.34 **(2)** 0.02 **(3)** 0.15 **(4)** 0.98

Q70. Match List-I with List-II.

List-I (Amine)	**List-II (Observation)**
(A) $C_6H_5NHCH_3$	(I) Liberates N_2 gas quantitatively with $NaNO_2$ and dil. HCl at 273–278 K
(B) $C_6H_5N(CH_3)_2$	(II) Does not react with benzenesulphonyl chloride
(C) $C_6H_5NH_2$	(III) Forms a diazonium salt, stable at 273–278 K, with $NaNO_2$ and dil. HCl
(D) $C_6H_5CH_2NH_2$	(IV) Gives with benzenesulphonyl chloride a product insoluble in aqueous KOH

Choose the correct answer from the options given below:

(1) (A)-(II), (B)-(IV), (C)-(III), (D)-(I) **(2)** (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

(3) (A)-(IV), (B)-(II), (C)-(III), (D)-(I) **(4)** (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

CHEMISTRY – SECTION B (Q71–Q75) Numerical value (integer answer)

Q71. For a reaction, the equilibrium constant is $K = 1.0 \times 10^{-2}$ at 500 K and the standard entropy change is $\Delta S^\circ = +40.0$ J K^{-1} mol^{-1}. The standard enthalpy change ΔH° of the reaction at 500 K is _____ kJ mol^{-1} (nearest integer).

(Given: $R = 8.314$ J K^{-1} mol^{-1}, $\ln 10 = 2.303$)

Q72. At 350 K, pure N_2O_4 is allowed to attain the equilibrium $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ in a closed vessel. At equilibrium, the total pressure is 3 atm and N_2O_4 is 50% dissociated. If the standard enthalpy change of the reaction is $\Delta_r H^\circ = +57.2$ kJ mol^{-1}, the standard entropy change $\Delta_r S^\circ$ of the reaction at 350 K is _____ J K^{-1} mol^{-1} (nearest integer).

(Given: $R = 8.314$ J K^{-1} mol^{-1}, $\log 2 = 0.30$, $\ln 10 = 2.303$. K_p is expressed in atm.)

Q73. The energy required to convert 6.0 mg of gaseous magnesium atoms completely into $Mg^{2+}(g)$ ions is _____ J.

(Given: $\Delta_i H_1$ and $\Delta_i H_2$ of Mg are 737 and 1451 kJ mol^{-1} respectively; molar mass of Mg = 24 g mol^{-1})

Q74. 14.8 g of propanoic acid is converted into its acid chloride with $SOCl_2$ and then into the amide B with excess NH_3. B on heating with Br_2 and aqueous NaOH gives the amine C. Assuming quantitative conversion in each step, the mass of C obtained is _____ g.

(Given: atomic masses C = 12, H = 1, N = 14, O = 16)

Q75. Acetanilide on treatment with Br_2 in acetic acid gives the major product Q, which on hydrolysis with aqueous NaOH gives R. In Kjeldahl's method, the ammonia evolved from 1.72 g of R is completely neutralised by V mL of 0.1 M H_2SO_4. The value of V is _____.

(Given: atomic masses C = 12, H = 1, N = 14, O = 16, Br = 80)

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